Harsh Ledwani

+91-8510824637 | ledwani830@gmail.com | LinkedIn | GitHub | Portfolio

Education

Symbiosis Institute of Technology, Pune — Expected May 2027
B.Tech – Computer Science Engineering

Greenway Modern Sr. Sec. School, New Delhi — April 2023
Class XII – CBSE

Technical Skills

Languages: C++, Python, Java, JavaScript, C, SQL
Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Responsive Design
Backend: Node.js, Express.js, FastAPI, REST API Design, API Integration, Microservices
Databases: MongoDB, MySQL, PostgreSQL, Supabase, Redis
Cloud & DevOps: AWS (EC2), Docker, CI/CD Pipelines, Git, GitHub, Linux
ML / AI: Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Data Modeling
Core CS: Data Structures & Algorithms, Object-Oriented Programming, System Design, Operating Systems, Computer Networks, Distributed Systems, DBMS

Projects

F1 Race Analytics Platform | *React.js, Node.js, REST APIs, Data Visualization* | Live | GitHub

- Engineered a full-stack Formula 1 analytics application with interactive dashboards visualizing driver performance, lap times, pit stop strategies, and season standings across multiple seasons.
- Integrated multiple external REST APIs and structured historical datasets to enable real-time data processing and deep historical analysis with sub-second response times.
- Architected a scalable backend with modular REST endpoints and optimized frontend rendering for cross-device performance and responsive UI.

SecureFlow – Fraud Detection System | *Python, ML, Blockchain, FastAPI, Redis* | Live | GitHub

- Developed a machine learning–based fraud detection system for UPI transactions using risk-based authentication, anomaly detection algorithms, and behavioral analysis pipelines.
- Implemented a FastAPI backend with Redis caching for real-time transaction scoring, reducing detection latency and enabling concurrent request handling at scale.
- Published peer-reviewed research paper on this system in the *International Journal of Aquatic Research and Environmental Studies* (ISSN: 2980-7840).

Neuro-Adaptive AI Learning Assistant | *Python, React.js, AI/ML, REST APIs* | GitHub — Ongoing

- Building an AI-powered adaptive learning platform that personalizes curriculum difficulty in real time using interaction pattern analysis, response accuracy tracking, and session history.
- Implementing a dynamic assessment engine with knowledge-state tracking and learning path optimization using cognitive load theory, along with backend logic for behavioral analysis and intelligent feedback generation.

Publications

"SecureFlow: Detecting Fraudulent Payments in UPI Transactions through Blockchain

- **and Machine Learning Models"** – Accepted and published in the *International Journal of Aquatic Research and Environmental Studies* (Electronic ISSN: 2980-7840), 2025. Peer-reviewed research on ML-driven anomaly detection and blockchain-based verification for securing digital payment ecosystems.

Leadership & Activities

Technical Workshop Instructor | *C Programming, Linux Fundamentals*

- Conducted workshops on C programming and Linux fundamentals for 50+ junior peers, strengthening foundational programming and command-line proficiency.

Robotics Workshop Organizer & Technical Fest Design Team

- Organized robotics workshops for school students; contributed to college technical fest planning, creative direction, and event execution.